MarketMind AI - Forecasting in Milestone 2

Three forecast experiences, each trained for its own business question and protected by its own role scope.

1. Forecasts implemented

Business Owner - Revenue Forecast

Selected model: Weekday-aware Linear Trend

Target: Daily net revenue in INR for the next 7, 14 or 30 days

Dataset: Amazon sales records with order date, Amount, category and status

Why this model: On the short history, recent trend plus weekday pattern produced the lowest MAE among five candidates.

Dashboard output: Revenue total, growth, chart, interval, comparison and CSV export.

MAE 77,902	RMSE 107,282	~25% lower MAE

Sales Executive - Personal Sales Forecast

Selected model: Seasonal Naive

Target: The signed-in seller's daily net sales for 7, 14 or 30 days

Dataset: Valid processed INR sales and returns with seller assignment

Why this model: The small history favoured its recent weekly pattern; a complex model was not forced onto limited data.

Dashboard output: Personal total, trend, chart and table under the authenticated seller scope.

207 source rows	MAE 748	Seller-scoped

Store Manager - Product Demand Forecast

Selected model: XGBoost Regressor

Target: Daily store-product demand for the next 7, 14 or 30 days

Dataset: Train/eval Parquet history with lag, calendar, discount, activity, stock and weather fields

Why this model: XGBoost best captured interacting lag, calendar and context effects in chronological evaluation.

Dashboard output: Product trends, filters, mapping status and stock-risk context for the assigned store.

250 series	MAE 2.152	R2 0.918

Administrator - Model Monitoring

Selected model: Monitoring view, not a fourth prediction model

Target: Forecast engine status, model versions, metrics and import jobs

Dataset: Versioned forecast_model_runs, forecast_predictions and forecast_jobs records

Why this model: Administrators need tenant-bound visibility across business, store and personal forecast scopes.

Dashboard output: Engine health, current model, horizons, model registry and recent jobs after MFA.

MFA required	All model scopes	Tenant-bound

How Forecasting Moves from Data to Dashboard

Model training is an offline, repeatable pipeline; forecast reading is an online, secured API workflow.

2. Forecast development workflow

1. **Clean** — Validate dates, values, status and source fields
2. **Engineer** — Build lag, weekday, trend and context features
3. **Evaluate** — Chronological holdout; compare against baseline
4. **Select** — Lowest MAE, then RMSE; record all metrics
5. **Version** — Save predictions, report and model artifact
6. **Import** — Validate scope and upsert stable database keys
7. **Serve** — FastAPI checks role/scope and returns JSON
8. **Display** — React renders KPIs, charts, tables and insights

API	Allowed role/scope	Response used by frontend
GET /api/v1/forecasts/options	Role-specific report options	Supported horizons, categories and products
GET /api/v1/forecasts/revenue	Business Owner or Administrator	Algorithm, metrics, comparison, series, intervals and insights
GET /api/v1/forecasts/personal	Signed-in Executive; Admin-selected seller	Seller scope ID, personal series and model evidence
GET /api/v1/forecasts/demand	Assigned Store Manager; Admin-selected store	Product series, mapping status, inventory context and risk
GET /api/v1/forecasts/monitoring	Administrator after MFA	Model registry, engine health and forecast job status

What is working	All three forecast types are generated, versioned, imported, exposed through real APIs and shown in the correct role dashboard.
What must be stated honestly	Revenue has negative R2 because the dated history is short, so it is a functional prototype rather than production accuracy. Demand remains source_unit until sale_amount is confirmed. Unmapped products do not receive invented stock risk.
What comes later	Longer histories, confirmed data contracts, drift and matured-accuracy monitoring, scheduled retraining, production telemetry and formal model governance.